

Improving Explainable Chart Understanding in Vision-Language Models

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Abstract

Chart question answering remains challenging for Visual-Language Models (VLMs) because it requires both precise visual extraction and numerical reasoning. In this project, we investigate two resource-efficient techniques to improve performance without retraining the entire network: inference-time Chain-of-Thought (CoT) prompting and Parameter-Efficient Fine-Tuning via Low-Rank Adaptation (LoRA). We evaluate performance using exact match and qualitative error analysis on ChartQA dataset, with a focus on both answer accuracy and interpretability.

1. Introduction

Visual-Language Models (VLMs) perform strongly on many multimodal tasks, but chart question answering remains challenging because it requires precise reading of axes, legends, labels, and values together with numerical reasoning. Unlike natural images, charts encode structured quantitative information, so small perception errors can directly produce wrong answers.

In this project, we study two resource-efficient ways to improve chart understanding without retraining the full model: inference-time Chain-of-Thought (CoT) prompting and Low-Rank Adaptation (LoRA) fine-tuning. Using ChartQA, we compare three settings: a zero-shot baseline, a zero-shot CoT variant, and a LoRA-adapted model.

We evaluate these settings with exact match and qualitative error analysis to examine both accuracy and interpretability. Our goal is to understand when explicit reasoning helps, when lightweight adaptation helps, and how these choices affect reliable chart question answering.

2. Related Work

Chart question answering is more demanding than standard visual question answering because models must read chart structure, recover values, and perform numerical reasoning. Benchmarks such as DVQA, PlotQA, and ChartQA establish this challenge and highlight the need for chart-specific evaluation beyond generic visual recognition [2, 5, 4].

Recent instruction-tuned VLMs improve general multi-modal performance, but chart QA often still benefits from additional reasoning or task adaptation [3]. Chain-of-Thought prompting improves multi-step reasoning in language and multimodal settings [6, 7], while LoRA provides a parameter-efficient way to adapt large models with a small number of trainable updates [1]. Motivated by these directions, we compare zero-shot baseline, CoT prompting, and LoRA fine-tuning for chart understanding.

3. Methodology

We design an experimental pipeline for chart question answering with explainable reasoning.

Task setup. Given a chart image and a natural language question, the model predicts a short textual answer, such as a number, label, or phrase.

Dataset. We use the ChartQA dataset with standard train, validation, and test splits. The training split is used for adaptation, the validation split is used for development, and the test split is used for final evaluation.

Baseline. We first evaluate the pretrained Qwen2-VL-2B model in direct-answer mode. The model receives the chart image and question and is asked to return only the final answer.

CoT reasoning. We then add a reasoning-oriented prompt so that the model first identifies relevant chart structure and values, then performs the required comparison or computation before returning the final answer. Only the extracted final answer is used for evaluation.

Fine-tuning. We apply lightweight LoRA adaptation to the baseline model to assess whether fine-tuning improves chart-specific visual grounding and numerical reasoning.

Comparison settings. We compare:

- Baseline (direct answering) vs. CoT prompting
- Baseline (Zero-shot) vs. LoRA fine-tuned model
- Accuracy gains vs. interpretability gains

4. Experimental Settings

We conduct experiments under a unified evaluation setup to ensure fair comparison across all configurations.

Model configuration. We use Qwen2-VL-2B-Instruct as the backbone model in all experiments. The baseline, CoT, and adapted settings share the same model family so that the comparison focuses on the effect of direct answering, reasoning-oriented prompting, and lightweight adaptation.

Data usage. The training split is used for lightweight adaptation, the validation split is used for prompt development and intermediate comparison, and the test split is reserved for final evaluation. This separation ensures that model selection and prompt refinement do not directly affect the final reported results.

Inference settings. All experiments are run with the same decoding and preprocessing pipeline whenever possible. The baseline setting uses shorter responses, while the CoT setting allows longer generation to accommodate intermediate reasoning. For stable inference in our compute environment, experiments are run with batch size 1.

Reasoning setting. In the CoT configuration, the model is prompted to produce intermediate reasoning before the final answer. To reduce unnecessary reasoning on simple questions, we also test a routed setting in which only harder question types, such as counting, aggregation, comparison, and arithmetic reasoning, are assigned to the CoT prompt.

Implementation details. All experiments are implemented in Python with PyTorch and Hugging Face Transformers. The Qwen2-VL processor is used to prepare image-text inputs. Inference is run with batch size 1. The CoT setting uses a longer generation length than the baseline to allow reasoning steps, and generated outputs are post-processed to extract the final short answer.

LoRA setting. For lightweight adaptation, we use LoRA adapters with rank $r=8$, $\alpha=16$, dropout 0.05, and no bias term. The adapters are applied to the transformer projection layers q_{proj} , k_{proj} , v_{proj} , and o_{proj} , while the vision tower is frozen. This results in about 2.18M trainable parameters, or roughly 0.985% of the full model. We train for 1 epoch with batch size 1, gradient accumulation 8, learning rate 2×10^{-4} , maximum sequence length 1024, and gradient checkpointing enabled.

Evaluation protocol. We report Exact Match (EM) as the main metric. Beyond overall accuracy, we compare the models qualitatively through success and failure cases to examine stability, reasoning ability, and output interpretability.

5. Results

Overall accuracy. On the ChartQA test set, the baseline model achieves **47.76%** accuracy, the CoT model achieves **45.68%**, and the LoRA-adapted model achieves **52.08%**. Among the three settings, LoRA provides the strongest overall performance, improving over the baseline by **3.92** EM points.

Head-to-head comparison for CoT. On the test set, both models answer **544** examples correctly. The baseline model correctly answers **53** additional examples that the CoT model misses, while the CoT model correctly answers **27** examples that the baseline gets wrong. Both models fail on **626** examples.

These results show that the baseline remains more reliable overall, although the CoT model still provides some complementary improvements on a smaller subset of questions.

Head-to-head comparison for LoRA. On the test set, the baseline and LoRA models both answer **560** examples correctly. The baseline model correctly answers **37** additional examples that LoRA misses, while the LoRA model correctly answers **86** examples that the baseline gets wrong. Both models fail on **567** examples.

These results show that LoRA not only improves the overall EM score, but also solves substantially more questions beyond the baseline, giving a net gain of **49** correct answers.

6. Analysis and Discussion

6.1. Performance Analysis

Baseline model. The main strength of the baseline model is stability. Its answers are usually short, direct, and consistent with the question format. It performs especially well on direct lookup, precise numeric reading, simple counting, and straightforward comparison questions. Because it does not generate long intermediate reasoning, it is less likely to drift away from the target answer or introduce extra formatting errors.

However, its weakness appears on multi-step questions. When the model needs to combine several chart values, compute a sum or median, or reason across multiple points, it often fails because the answer cannot be obtained from a single direct lookup.

CoT model. The main advantage of the CoT model is its potential on multi-step reasoning. It performs better on some questions involving summation, threshold comparison, and counting across multiple points. In addition, CoT provides a more explainable direction, since reasoning steps can make the model’s prediction easier to interpret.

However, the output of CoT model becomes more complicated and less stable. In many cases, the reasoning process introduces new mistakes, such as wrong intermediate values, incorrect arithmetic, or a final answer that does not match the intended format. This suggests that the current CoT design has potential, but its prompt and answer control are not yet strong enough.

LoRA model. The main advantage of the LoRA model is better chart-specific grounding. Compared with the baseline, LoRA improves both numeric and text-or-label predic-

tions, as illustrated by the examples in Figure 2. Qualitatively, LoRA more often corrects errors that require precise chart reading, such as selecting the correct year, category, series, or exact segment value.

However, LoRA shows weaker answer-format stability in some cases. Under strict exact-match evaluation, it can lose points when it outputs extra characters, multiple answers, or strings that are semantically correct but not normalized to the expected format. This suggests that stronger answer normalization or tighter answer-only constraints could further improve measured EM.

Project resources. The code, outputs, and supplementary results for this project are available in the Google Drive folder.

6.2. Case Examples

Baseline vs. CoT

We present four representative examples from the test set in Figures 1. The first two cases show situations where the baseline model gives more stable and accurate answers, while the last two cases show situations where CoT is more helpful for multi-step reasoning.

Baseline vs. LoRA

We present four representative examples from the test set in Figure 2. The first two cases show situations where the baseline model is more stable under strict exact-match evaluation, while the last two cases show situations where LoRA better grounds visual structure and predicts the correct chart-specific answer.

6.3. Discussion and Potential Improvement

While our current experiments isolate the individual impacts of inference-time CoT prompting and lightweight LoRA fine-tuning, there are two highly promising avenues for future improvement. First, the CoT approach could be enhanced by implementing a two-stage architecture, decoupling the rationale generation from the final answer prediction to minimize strict-formatting conflicts. Second, because the ChartQA dataset currently lacks ground-truth reasoning steps, our LoRA adapters were trained strictly on direct answers. Should "golden" reasoning annotations become available, future research could apply LoRA to directly fine-tune the reasoning pathways themselves (CoT distillation). This would effectively combine the parameter efficiency of LoRA with the logical interpretability of CoT.

7. Conclusion

This project investigates two resource-efficient strategies for improving chart question answering in Vision-Language Models: inference-time Chain-of-Thought (CoT) prompting and parameter-efficient fine-tuning with LoRA. Our ex-

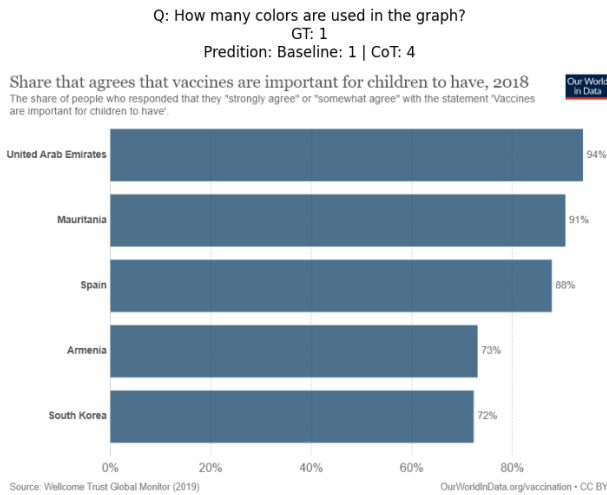
periments on ChartQA show that LoRA is the most effective setting overall, achieving **52.08%** exact-match accuracy, compared with **47.76%** for the baseline and **45.68%** for CoT. In head-to-head comparison, LoRA also solves substantially more examples beyond the baseline, indicating that lightweight adaptation improves chart-specific visual grounding and answer accuracy.

At the same time, the results show that CoT remains useful for interpretability and for some multi-step questions, such as counting, aggregation, and threshold comparison. However, its overall performance is limited by unstable intermediate reasoning and answer-format errors. These findings suggest that, for chart understanding, lightweight adaptation currently provides a more reliable gain than prompting alone, while reasoning-based methods remain valuable for explaining model behavior.

Overall, our study shows that both adaptation and reasoning matter, but they contribute in different ways. LoRA improves domain-specific grounding and overall accuracy, whereas CoT improves transparency and can help on selected reasoning-heavy cases. Future work can combine these strengths through better answer normalization, two-stage reasoning-and-answer pipelines, and, when available, supervision for high-quality reasoning traces.

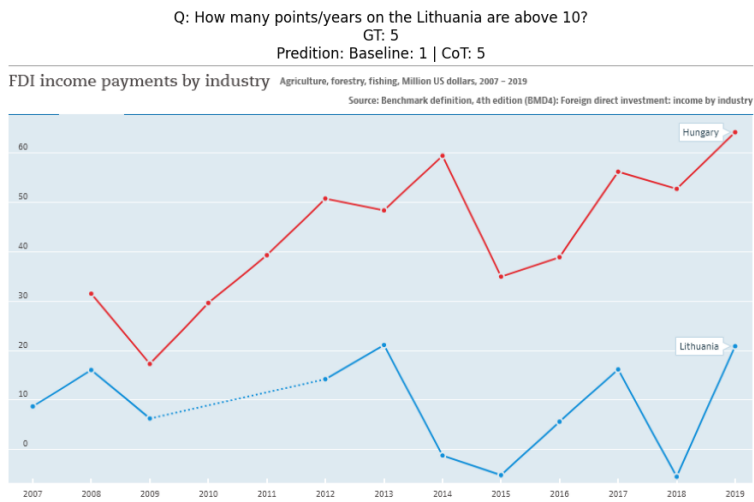
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(a) Baseline success. Baseline: 1; CoT: 4.

The baseline is more stable on simple lookup questions.



(c) CoT success. CoT: baseline: 1; 5.

CoT helps when multiple points must be counted.

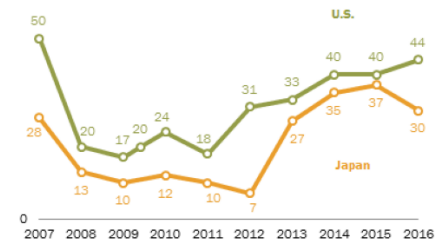
Figure 1. Representative case examples comparing baseline and CoT performance. The top row shows cases where the baseline is more stable and accurate, while the bottom row shows cases where CoT is more effective for multi-step reasoning.

Q: What is the median value of Japan graph from 2013 to 2015?
GT: 35
Prediction: Baseline: 35 | CoT: 30

Ratings for U.S. economy continue post-crisis climb, but views of Japan's economy sputter

The current economic situation in our country is good

70%



Source: Spring 2016 Global Attitudes Survey, Q3.

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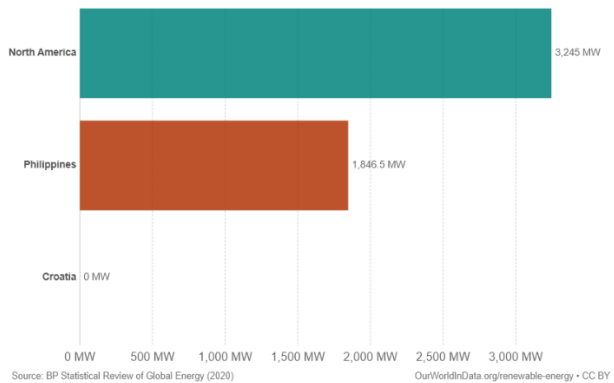
(b) Baseline success. Baseline: 35; CoT: 30.

The baseline is more reliable for exact numeric reading.

Q: Is the average value of Philippines and North America greater than 2500 MW?
GT: Yes
Prediction: Baseline: No | CoT: Yes

Installed geothermal energy capacity, 2005

Cumulative installed capacity of geothermal energy, measured in megawatts.



(d) CoT success. baseline: No; CoT: Yes.

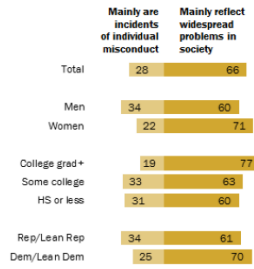
CoT helps with multi-step threshold reasoning.

Q: What value is been shown twice in the Mainly reflect widespread problems in society?

GT: 60
Prediction: Baseline: 60 | LoRA: [60, 70]

Most Americans say reports of sexual misconduct reflect societal problems

% who say recent allegations of sexual harassment and assault ...



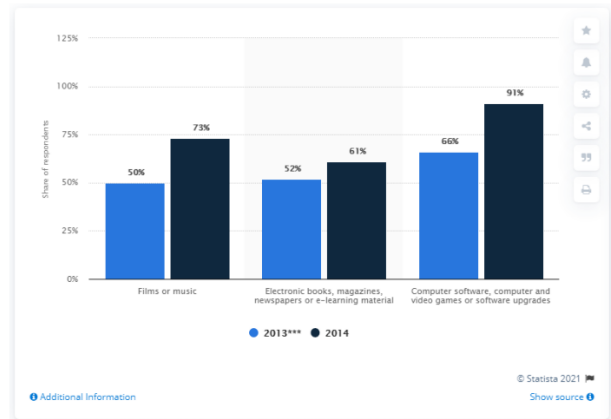
Note: Don't know responses not shown.
Source: Survey of U.S. adults conducted Nov. 29-Dec. 4, 2017.
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(a) Baseline success. Baseline: 60; LoRA: [60, 70].

LoRA identifies the relevant values but returns multiple answers, so the baseline is better aligned with strict exact-match evaluation.

Q: Which year has lowest Share of individuals who downloaded purchased media online in Great Britain in 2013 and 2014, by type of media?

GT: 2013
Prediction: Baseline: 2013 | LoRA: 2013***



(b) Baseline success. Baseline: 2013; LoRA: 2013**.

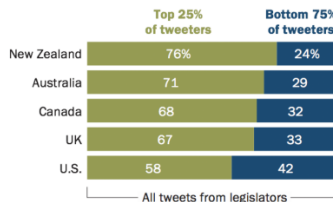
The baseline is more reliable when the task requires a clean, exactly formatted final answer.

Q: Find missing data of the sequence 24, _, 32, 33, 42?

GT: 29
Prediction: Baseline: 32 | LoRA: 29

A subset of legislators dominates the Twitter conversation

% of all tweets from legislators created by the ...



Source: Analysis of tweets from national-level legislators in the United States, United Kingdom, Canada, Australia and New Zealand, posted Jan. 1-June 30, 2019. N=2,180 legislators with Twitter accounts, including 2,056 who tweeted at least once.
*For Global Legislators on Twitter, an Engaged Minority Creates Outsize Share of Content"
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(c) LoRA success. Baseline: 32; LoRA: 29.

LoRA better grounds the chart structure and selects the correct value instead of a nearby distractor.

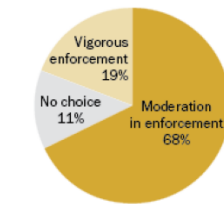
Figure 2. Representative case examples comparing baseline and LoRA performance. The top row shows baseline wins caused by stronger answer-format stability, while the bottom row shows LoRA wins driven by better chart-specific grounding and label/value selection.

Q: Which colored segment is the tiniest?

GT: Gray
Prediction: Baseline: Vigorous enforcement | LoRA: gray

Moderate Enforcement of 1964 Law Much Preferred

Percent who prefer ... of the new civil rights law



Source: Opinion Research Corporation, Nov. 4-8, 1964
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(d) LoRA success. Baseline: Vigorous enforcement; LoRA: gray.

LoRA more accurately maps the visual cue in the chart to the correct target label.